

Software Developer — OCI Load Balancer Team, Oracle

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OCI's Load Balancer team is a distributed systems problem at global cloud scale. High-availability, correctness under concurrency, and automation are the engineering problems I find most interesting — and where my work is most directly applicable.

Why I am a strong fit:

- **Distributed systems at load-tested scale.** AlgoPulse: Temporal.io workflows, Kafka event streaming, Redis pub/sub live leaderboard, deployed on AWS with auto-scaling groups, k6 load tested to 10,000 concurrent users with sub-100ms leaderboard latency guaranteed at 2,000 req/sec. KubeServe: KEDA autoscaler (scale-to-zero to N replicas), circuit breaker with EWMA load balancing, k6 tested to 500 VUs. These are not toy systems — they are designed to hold up under real load.
- **Java correctness under concurrency.** InventoryReservationService: Java 21 + Spring Boot REST API with PostgreSQL SELECT FOR UPDATE row-level locking across multi-SKU reservations, idempotent POST endpoints with Flyway schema migrations, and Testcontainers concurrency tests that prove a qty-1 race produces exactly one success and one 409 Conflict. Clean, verifiable Java with correctness guarantees.
- **Operational engineering and automation.** At Isomer AI I owned 30+ per-tenant production deployments end-to-end — each deployed via ArgoCD GitOps, monitored with Grafana, and debugged across the full stack when issues arose. At Mphasis I built AWS and GCP CI/CD pipelines across teams. Reducing operational burden through tooling and automation is how I approach production systems.
- **Generalist who debugs across the stack.** At NC State I debug across genomic data pipelines, cloud infrastructure, and ML model pipelines simultaneously. At Capgemini I worked across Kafka, Spark, Hadoop, and NLP layers. The JD asks for engineers who can dive deep into any part of the stack — that is how I have always worked.

OCI's Load Balancer is infrastructure that the rest of the cloud depends on. I want to work on systems where correctness and scale are non-negotiable.

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